Calcium Amendment (Gypsum Replacement) — Technical Data

High-purity natural oolitic aragonite · Technical documentation

Technical Data

Calcium Amendment

Replacement option to Gypsum

Oolitic aragonite - the world's only recent origin, renewable and sustainable performance mineral

HI-CAL calcium amendment/Fine Granular (Aragonite)

Exceptionally Pure calcium carbonate

Another common calcium amendment is gypsum. Gypsum is different from oolitic aragonite in that it will not supply alkalinity to soil. In soils that require low pH with little buffering capacity it may be more effective. However, in situations where the addition of alkalinity to the soil is not a concern, a ton of aragonite will deliver 28% more calcium in comparison to a ton anhydrous gypsum and 60% more calcium than a ton hydrated gypsum.

Oolitic Aragonite is an extremely efficient source of calcium for agricultural and turf production. Its high solubility and surface area ensure that it will dissolve and deliver the calcium to the soil. At pH values between 5.5 and 6.8 aragonite is far more soluble than calcite and will thus require much less material to deliver the equivalent amount of calcium to the soil. Addition of aragonite to the soil will also result in the addition of alkalinity thus controlling pH by neutralizing acidic soils. Another added benefit is the natural buffering point of aragonite which lies at approximately 6.8. When aragonite is added to the soil the alkalinity will raise the pH of the soil as the $CaCO_3$ dissolves and reacts with acids in the soil, however when the pH starts to reach the buffering point, the $CaCO_3$ becomes less soluble and it will take very large amounts to raise the pH above the buffering point. In contrast to aragonite, calcite has a buffering point of approximately 5.5, thus the vast majority of the calcite added to the soil will not dissolve thus it will not effectively deliver calcium to the soil or raise the pH efficiently.

Aragonite Composition Large surface area and porosity

Properties

Oolitic Aragonite

Calcite/Limestone

Specific Gravity

2.93+

2.7

Mohs Hardness

3.5 - 4

3

Crystal Structure

orthorhombic

hexagonal

Surface Area

> 1.82m^2/g

< .55m^2/g

Zeta Potential

-33.85mV to - 6.65mV

-1.01mV to +11.55Mv

Micro porosity

Very high

Low

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